ANNUAL MEDICAL COST PREDICTOR

DATASCI347

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AGENDA

01 INTRODUCTION

02 APPROACH

03 SETUP

04 SELECTED DATA

05 EXPLORATORY ANALYSIS

06 METHODS

07 RESULTS

08 CONCLUSION

INTRODUCTION

What is the problem?

- US healthcare spending is the highest in the world, averaging $14,775 per person in 2024
- Medical costs vary widely between patients and can be unpredictable, creating financial uncertainty for many families
- The unpredictability makes it challegning for many individuals to plan their healthcare expenses and choose appropriate insurance plans

Why is this problem interesting?

- This problem impacts nearly all families and what insurance plan to get is often a big stressor for people
- Using this predictor, families can plan ahead of time and understand how their demographics, lifestyle, and healthcare utilization influence their projected medical costs, and therefore make decisions about their insurance plans and other finances

Sources: DA 2022, Axios 2024

INTRODUCTION

Our Project

- We aim to predict annual medical spending using patient-level data including demographics, lifestyle factors, pre-existing health conditions, and prior healthcare utilization

- We compare multiple models to evaluate how well different methods can estimate medical costs and understand what the key drivers of medical spending are so we can provide insights that helps individuals make more informed financial decisions

APPROACH

Linear Regression

- Common, baseline model used in many datasets
- Fits a linear model on the dataset with annual medical cost as the Y variable
- Can be useful for identifying model fit, statistical significance of variables, and understanding relationships between variables
- Has many limitations especially because the dataset includes complex non-linear relationships and interactions

Bagging

- Bagging is a good method for non-linear relationships and reduces variance
- Since insurance costs do not follow a linear pattern, bagging is a good approach for our data
- Key components include bootstrapping, parallel training, aggregation
- Hence its name: (B)ootstrap (Agg)regating

APPROACH (CONT.)

Random Forest	Gradient Boosting
• RF is optimal for non-linear relationships • Similar to bagging, it also reduces variance and overfitting, but also decorrelates trees • Random feature selection will be used at each split in the tree • Works well with the dataset because of complex relationships and variables and potential variable interactions	• Just like bagging and random forest, Gradient Boosting combines multiple decision trees • However, GB trees are dependent and sequential • Each tree builds off previous = reduce error • Slow to run (especially with tuning needs); prone to overfitting

SETUP

Target Variable

annual_medical_cost

Goal: Predict expected medical spending for a patient based on their demographics

Dataset includes 100,000 patient observations across 54 variables

Covers 4 main areas: demographics, past health conditions, lifestyle, utilization, insurance

Key Features

Demographics: age, income, bmi, education, sex, region, marital_status

Past Health: diabetes, asthma, had_major_procedure, cancer_history

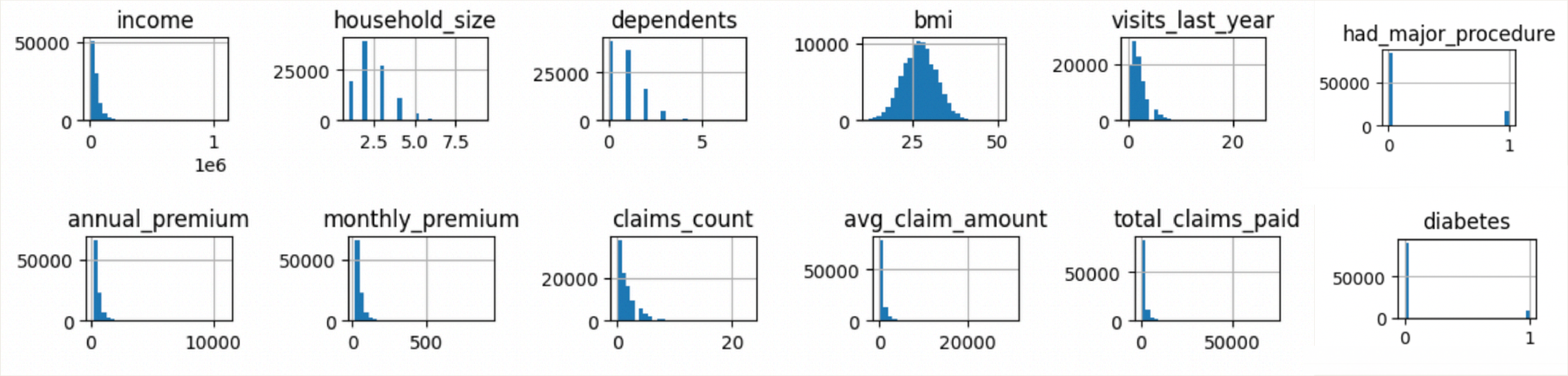
Lifestyle: alcohol_freq, smoker

Utilization: hospitilizations_last_3yrs, medication_count, claims_count

Insurance*: avg_claim_amount, monthly_premium, total_claims_paid*

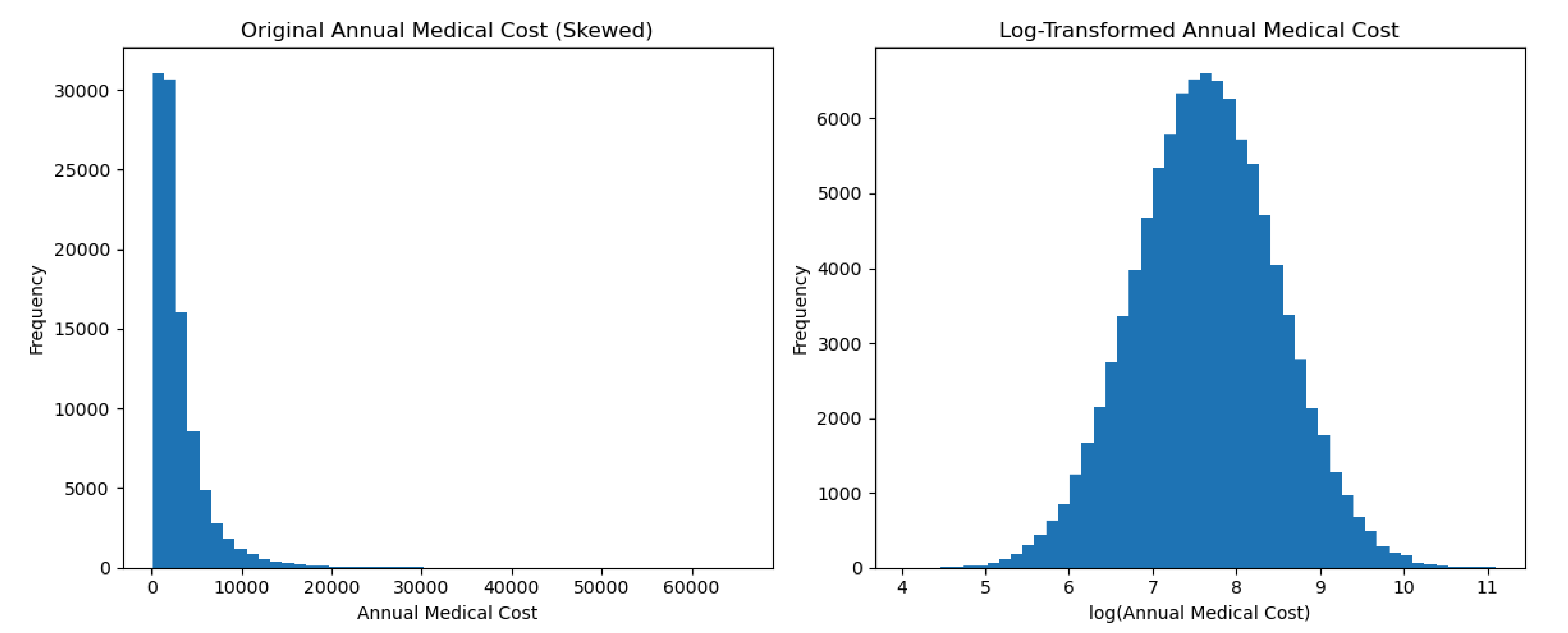
SETUP: EDA

Dataset Variable Distribution



- Most variables are right skewed
- Majority of individuals have incomes above $30,000 and small household sizes (~3)
- Past health conditions show that majority of patients are unaffected (diabetes, had_major_procedure)
- BMI only normally distributed variable
- Most have small number of claims with median claim amount at $657

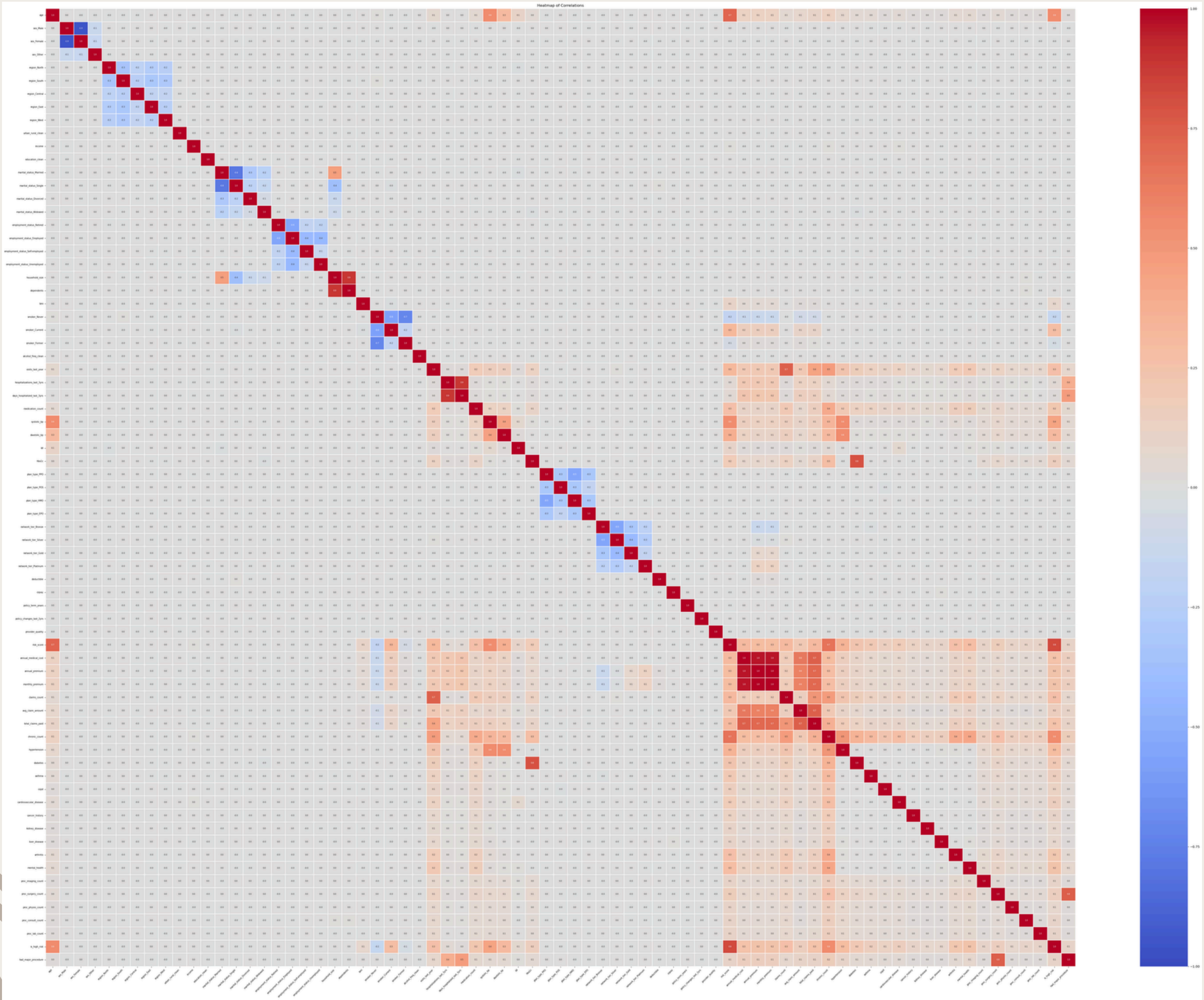
ANNUAL MEDICAL COST



The distribution of annual medical cost is highly right-skewed, so we applied a log transformation to evenly distribute the data, improving model stability and performance

np.log1p(cost)

SETUP: FEATURES



- Correlation matrix shows strong relationship between medical cost and other payment related variables (ie claim amount, monthly premium)
- Non-payment related variables show weak linear relation to medical cost
- Payment variables not best suited as predictors as they are too directly related and inaccessible in real-world settings

SETUP: METHOD

- **Evaluation Metrics**: R^2 & RMSE
- Personal computing environment on a Jupyter notebook with Python
- Variables selected using a heat map to determine which features were most correlated with medical cost and premium
    - Most did not have a strong corrrelation → not linearly related
    - Avoided using payment variables so that model would be more interpretable

Linear Regression

- Baseline method to see if there is a linear relationship between medical cost and other variables
- Does not require hyperparameters

Bagging

- Aggregated prediction from decision trees made by bootstrap samples of data
- Parameters:
    - Base model: DecisionTreeRegressor
    - n_estimators=200

Random Forest

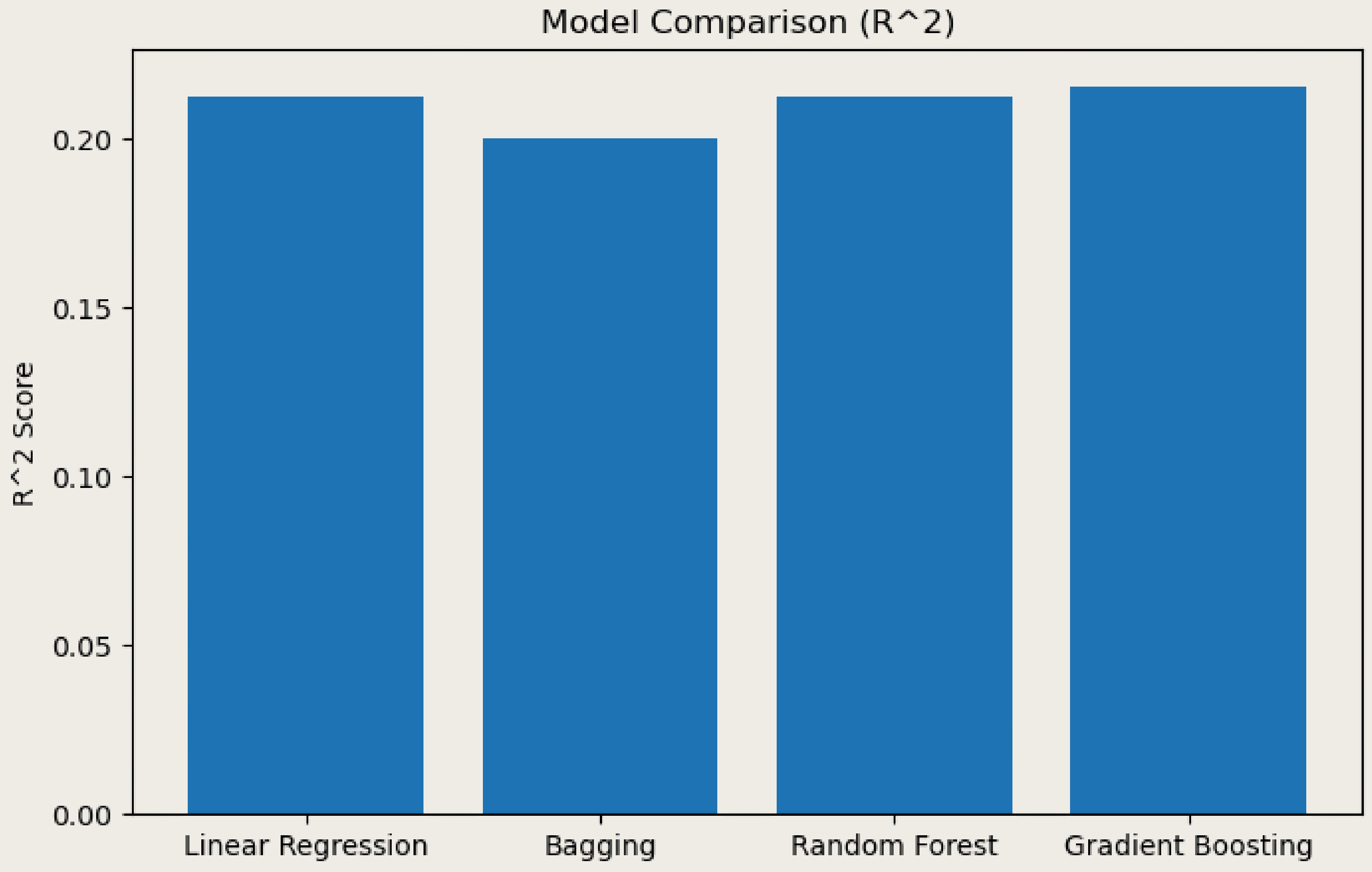
- Like bagging except random subsets of variables are chosen at each split
- Parameters:
    - n_estimators=200,
    - max_depth=10,
    - min_samples_leaf=5

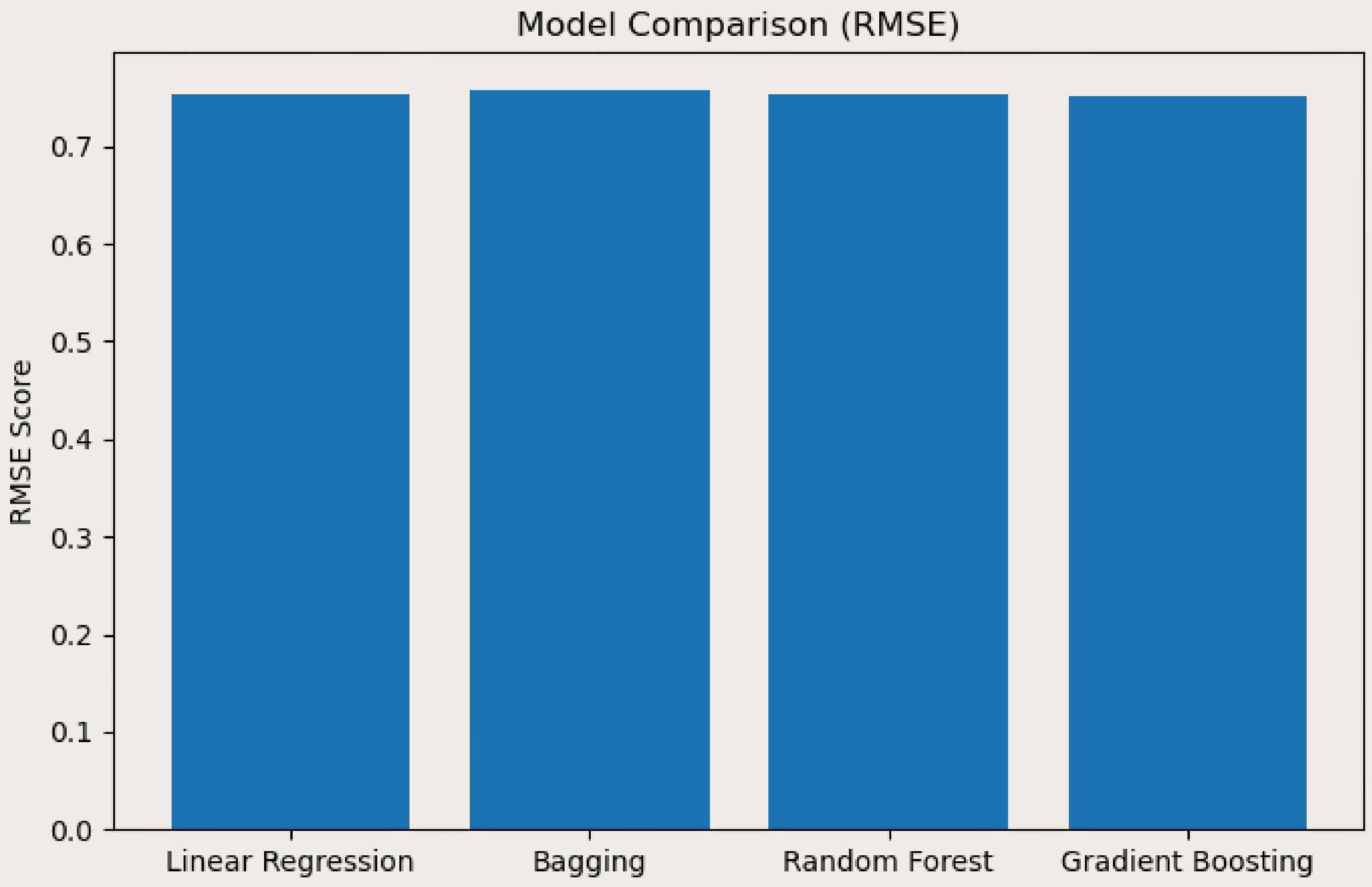
Gradient Boosting

- Each tree sequentially built to reduce error and final prediction is sum of trees
- Parameters:
    - n_estimators=200,
    - learning_rate=0.05,
    - max_depth=3,

RESULTS: METRICS

Model Comparison

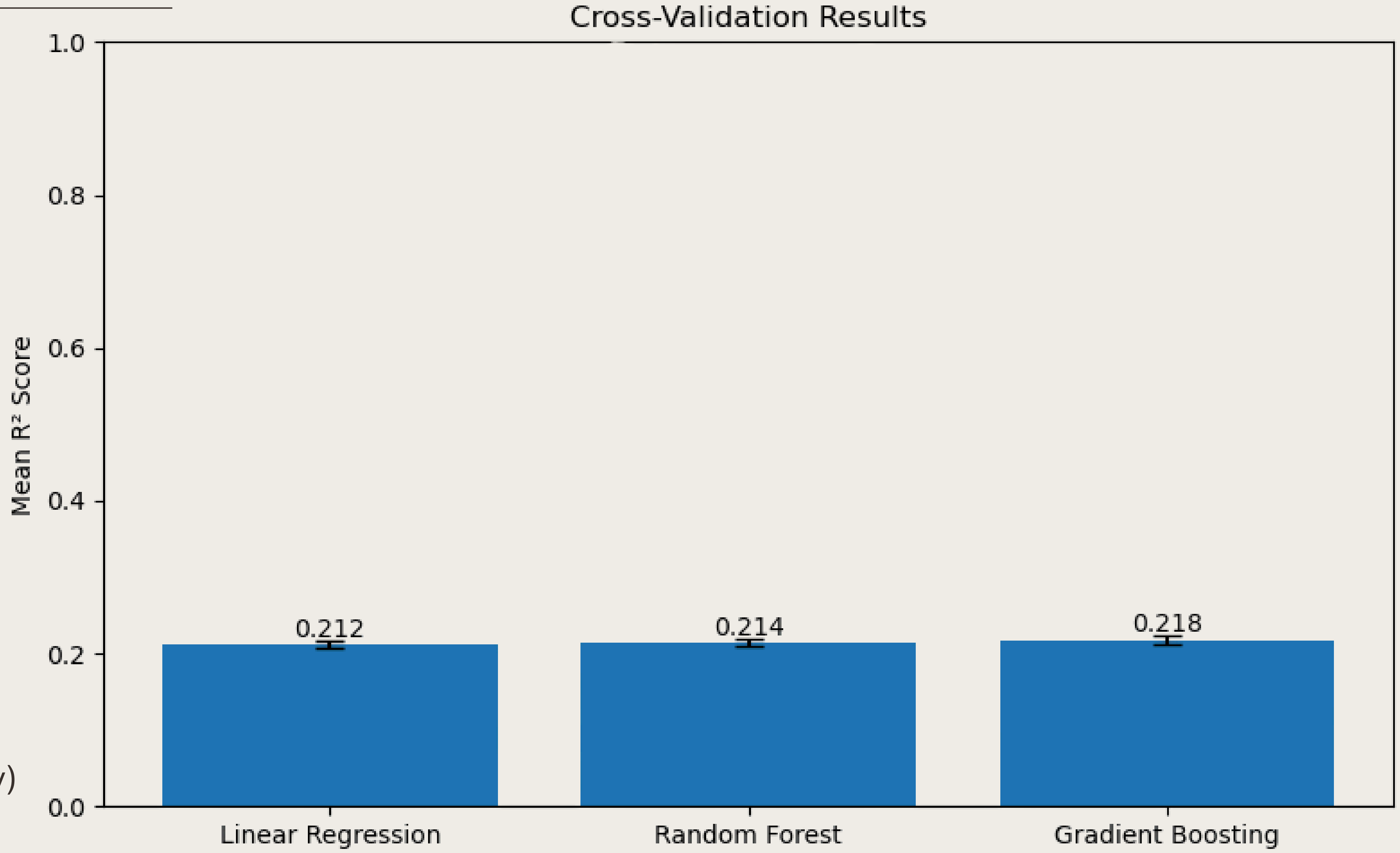




Linear regression, random forest, and gradient boosting perform very similarly for R^2 (~0.21) and RMSE (~0.75), with gradient boosting performing slightly better

RESULTS: CV

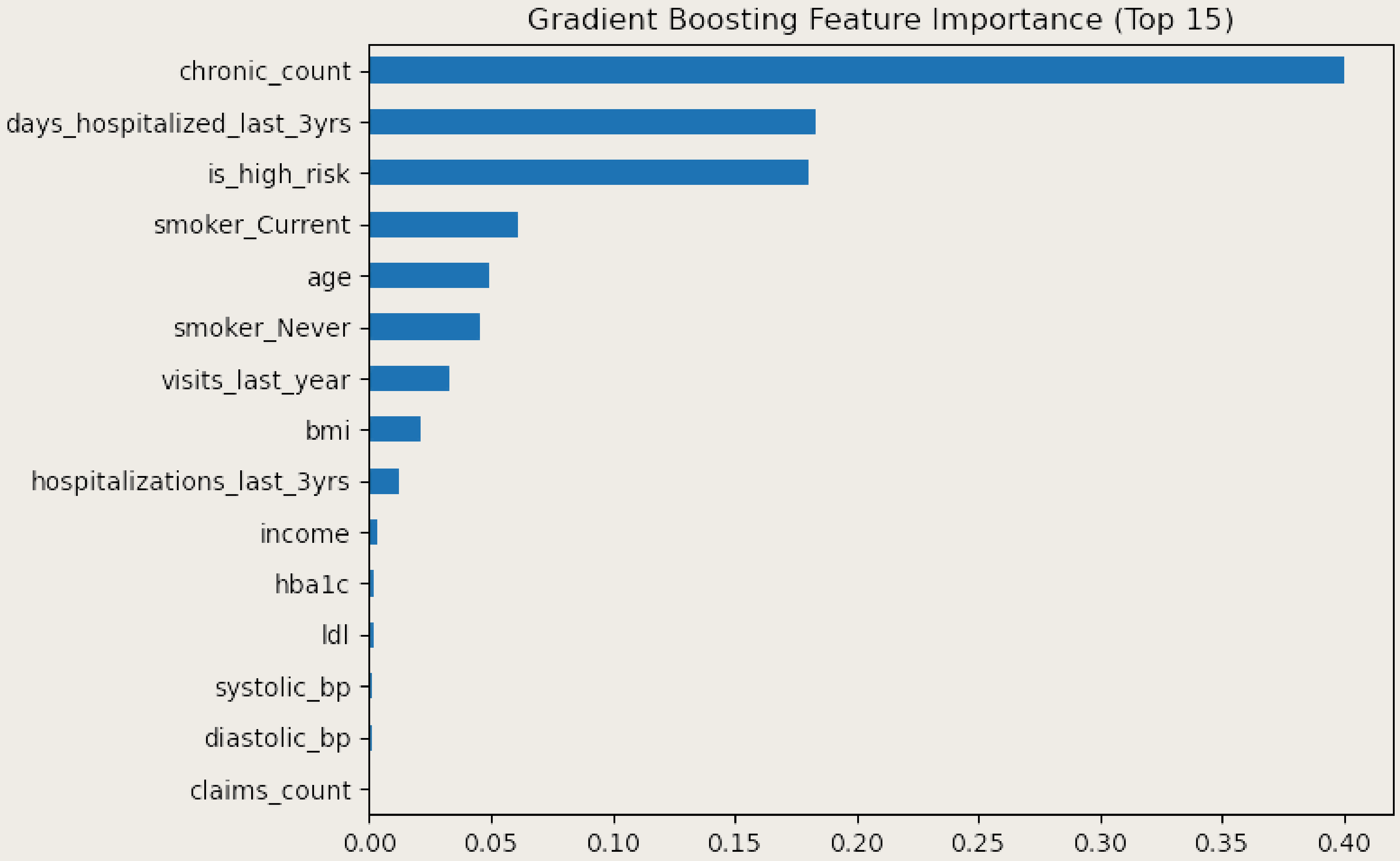
Cross Validation



Std: 0.004 - 0.006 (low variability)

The cross-validation results compare model performance using mean R^2 across five folds. Gradient Boosting achieves the highest avg. R^2 but all models perform similarly (~0.21). The small error bars indicate that performance is highly stable across different data splits

IMPORTANT FEATURES



For gradient boosting, the number of chronic conditions diagnosed is the most important feature, with number of days hospitalized also showing high importance

DISCUSSION

Low R^2 Across All Models
All of our models performed very similarly with a relatively low R^2 around 0.21

Reasons for Low Metrics

Poor Predictors:
- The variables included in the dataset may not be strong predictors of medical costs
- There may be variables not included that heavily influence costs, such as additional health factors, diet, and a lack of lifestyle variables

Dataset Limitations:
- Large dataset made it difficult to try permutations of different parameters
- Data collected from multiple sources and stitched together, not from one cumulative survey

Extrapolation Importance
- Despite having variables which increased our $R^{2,}$ we wanted our model to be used by people which had not yet incurred large direct healthcare expenses

INTERPRETATION

- Our results show that factors such as the number of chronic conditions someone has been diagnosed with, the number of days they have been hospitalized in the past year, and whether or not someone is a smoker plays an important role in determining in medical costs
- Individuals can use this model to better understand what their medical costs might look like based on their demographics and pre-exisiting health conditions
- This tool can be especially helpful for people who have been recently diagnosed with a condition and want to understand how their insurance costs/plans might change

THANK YOU!

REFERENCES

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https://pmc.ncbi.nlm.nih.gov/articles/PMC2838159/

Qualitative evidence backing up chronic illness as a strong predictor

ScienceDirect: Lifestyle Risk Factors Predict Healthcare Costs in an Aging Cohort
https://www.sciencedirect.com/science/article/abs/pii/S0749379705003004#:~:text=Lifestyle%20Risk%20Factors%20Predict%20Healthcare%20Costs%20in%20an%20Aging%20Cohort%20%2D%20ScienceDirect

Lifestyle risk factors affecting healthcare costs

GITHUB

[**Repository Link**]